

YUSHAN DENG

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EDUCATION

Shanghai Jiao Tong University

Sep. 2023 - Mar. 2026

Master of Engineering in Computer Science and Technology, 3.3/4.0 GPA

Research Focus: Computer Vision, Diffusion Models, Vision-Language Models, Medical Image Processing

Sun Yat-sen University

Sep. 2019 - Jun. 2023

Bachelor of Engineering in Intelligent Science and Technology, 3.5/4.0 GPA

Main courses: Data Structures and Algorithms, Operating Systems, Embedded Systems and Design, Computer Vision, Natural Language Processing, Internet of Things Technology and Applications, Deep Learning, Software Engineering, Computer Networks

PUBLICATIONS

- [1] **Deng, Yushan** and Jiang, Yuqiu and Hong, Yi, “DiffLGE: Diffusion-Based Image Generation for Synthesizing Late Gadolinium Enhancement Scans,” *2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2024.
- [2] Ye, Huping* and **Deng, Yushan*** and Hong, Yi, “Epicardial Adipose Tissue Segmentation in MRIs Using Text-Prompted Pretraining Model,” *2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2024.
- [3] Qiuhui Chen and **Yushan Deng** and Xuancheng Yao and Yi Hong, “AD-Reasoning: Multimodal Guideline-Guided Reasoning for Alzheimer’s Disease Diagnosis,” *2026 IEEE International Conference on Multimedia and Expo (ICME)*, 2026.
- [4] Chaofan Lei, Jun Li, **Yushan Deng** and Xiaojun Tan, “RRT*ASV: Improved RRT* Path Planning Method for Ackerman Steering Vehicles,” *Expert Systems With Applications*, 2025, Art. no. 127349.

PROJECTS

Diffusion-Based Image Generation for Synthesizing Late Gadolinium Enhancement Scans [1]

2024

- Proposed DiffLGE, a novel approach that synthesizes high-quality LGE CMR images without gadolinium-based contrast agents by leveraging diffusion models.
- Conditioned the generative network on multimodal inputs (CINE and T1 mapping data) within a stable diffusion framework.

Epicardial Adipose Tissue Segmentation in MRIs Using Text-Prompted Pretraining Model [2]

2024

- Developed a vision-language pretraining model by adapting the Segment Anything Model (SAM) with text prompts, tailored for accurate segmentation of epicardial adipose tissue (EAT) in cardiac MRIs.
- Leveraged multiple public datasets for pretraining and introduced pseudo-labeling for data augmentation, culminating in fine-tuning on a private dataset to address data scarcity issues.

AD-Reasoning: Guideline-Guided Multimodal AD Diagnosis [3]

2026

- Built AD-MultiSense, a 10,378-visit multimodal AD diagnostic QA dataset integrating structural MRI with six clinical modalities and guideline-validated rationales.
- Developed AD-Reasoning with bidirectional cross-attention fusion, LoRA tuning, and GRPO verifiable rewards, reaching 93.33% / 92.82% ACC on CN-vs-CI and CN-vs-MCI.

agent-trading: Multi-agent Trading Research Framework

2026

- Built a Python framework for multi-agent trading research, integrating rule/factor engines, backtesting, broker-adaptor boundaries, JSONL audit logs, and a local FastAPI dashboard.
- Designed conservative safety defaults with preview-only execution, disabled live submission/cancellation, sanitized examples, CI tests, and open-source reproducibility docs.

3D Medical Image Report Generation Based on Vision-Language Mixture of Expert Model 2025

- Combined vision-language large models to automatically generate clinical reports for 3D medical images, improving report generation efficiency.
- Optimized the pipeline for 3D medical images and incorporated a MoE models to enhance generation quality.

Path Planning for Vehicles Based on Improved RRT* Algorithm [4] 2023

Undergraduate Thesis

- Integrated Ackermann steering model kinematic constraints to alleviate control layer pressure.
- Employed Dijkstra's algorithm on road network topology models to compute shortest paths, reducing global path planning pressure.
- Optimized RRT* search space and pruned branches to accelerate algorithm convergence.

Intelligent Vehicle Technology Verification Platform 2022

Guangdong Provincial Department of Science and Technology 2020 Major Project

- Built a motion capture and positioning system, deployed the intelligent vehicle embedded system, debugged the intelligent vehicle hardware, and designed the intelligent vehicle line-following algorithm.

EXPERIENCE

Zelos (Suzhou) Technology Co., Ltd., Algorithm Engineer Aug. 2025 - Oct. 2025

- Contributed to multimodal 3D object detection and tracking for autonomous driving, covering requirements analysis, literature review, and technical design around Sparse4D, CenterPoint, PF-Track, and TransFusion.
- Debugged and validated a point-cloud/image fusion framework covering range alignment, BEV-query fusion, multi-sweep aggregation, and MV2D/MV2DFusion integration; benchmarked the results against the PF-Track tracking baseline.

Guangzhou Zhiwei Technology Co., Ltd., Algorithm Engineer Jul. 2022 - Aug. 2022

- Developed embedded systems for intelligent vehicles with a focus on designing robust path tracking and control algorithms.
- Engineered a dual-lookahead pure pursuit algorithm to enhance vehicle path tracking, resulting in improved turning stability.
- Devised a PID-based front wheel steering control method to mitigate virtual position errors caused by mechanical backlash.

CAMPUS EXPERIENCE

Shanghai Jiao Tong University, Computer Vision, Teaching Assistant Feb. 2025 - Jul. 2025

Sun Yat-sen University, ACM Team, Competitor Sep. 2019 - Jul. 2022

- Sun Yat-sen University Programming Contest 2020 Shenzhen Campus Leader
- Sun Yat-sen University Programming Contest Novice Contest 2021 Shenzhen Campus Leader

Sun Yat-sen University, School of Intelligent Engineering, Organizer Sep. 2020 - Jul. 2021

Sun Yat-sen University, Baseball Team, Shenzhen Campus Leader Sep. 2020 - Jul. 2021

Sun Yat-sen University, will Anime Club, Shenzhen Campus Core Member Sep. 2020 - Jul. 2022

SKILLS

- **Programming:** Python, C++, JavaScript, MATLAB, C#, $\text{\LaTeX}$
- **Languages:** Mandarin, English (TOEFL 4.5/6.0), Cantonese, Japanese
- **Other:** Git, Docker